

指令飞行

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Event

flyto 执行结果事件通知

Topic: thing/product/{gateway_sn}/events

Direction: up

Method: fly_to_point_progress

Data:

Column	Name	Type	constraint	Des
fly_to_id	飞向目标点 ID	text		
status	状态	enum_string	{ "wayline_cancel": "取消飞向目标点", "wayline_failed": "执行失败", "wayline_ok": "执行成功，已飞向目标 }	

			行中"}	
result	返回码	int		非0表示错误
way_point_index	当前执行到第几个航点	int		
remaining_distance	剩余任务距离	float	{"step":0.1,"unit_name":"米 / m"}	
remaining_time	剩余任务时间	float	{"step":0.1,"unit_name":"秒 / s"}	
planned_path_points	规划的轨迹点列表	array	{"size": -, "item_type": struct}	
»latitude	轨迹点纬度	double	{"max":90,"min":-90}	轨迹度, 值, 负, 正, 小数位
»longitude	轨迹点经度	double	{"max":180,"min":-180}	轨迹度, 值, 正, 负, 小数位
»height	轨迹点高度	float	{"step":0.1,"unit_name":"米 / m"}	轨迹度,

Example:

```

    "data": {
      "fly_to_id": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxx",
      "planned_path_points": [
        {
          "height": 123.234,
          "latitude": 13.23,
          "longitude": 123.234
        }
      ],
      "remaining_distance": 0,
      "remaining_time": 0,
      "result": 0,
      "status": "wayline_progress",
      "way_point_index": 0
    },
    "need_reply": 1,
    "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxx",
    "timestamp": 16540709686556,
    "method": "fly_to_point_progress"
  }
}

```

一键起飞结果事件通知

Topic: thing/product/{gateway_sn}/events

Direction: up

Method: takeoff_to_point_progress

Data:

Column	Name	Type	constraint	Des
status	任务状态	enum_string	{"task_finish":"一键起飞任务完成","task_ready":"准备起飞","wayline_cancel":"取消飞向目标点","wayline_failed":"执行失败","wayline_ok":"执行成功，已飞向目标点","wayline_progress":"执行中"}	
result	返回码	int		非0为错误

flight_id	一键起飞任务UUID	text		
track_id	航迹ID	text		
way_point_index	当前执行到第几个航点	int		
remaining_distance	剩余任务距离	float	{"step":0.1,"unit_name":"米 / m"}	
remaining_time	剩余任务时间	float	{"step":0.1,"unit_name":"秒 / s"}	
planned_path_points	规划的轨迹点列表	array	{"size": -, "item_type": struct}	
»latitude	轨迹点纬度	double	{"max":90,"min":-90}	轨迹度, 值, 负, 正, 小数位
»longitude	轨迹点经度	double	{"max":180,"min":-180}	轨迹度, 值, 正, 负, 小数位
»height	轨迹点高度	float	{"step":0.1,"unit_name":"米 / m"}	轨迹度,

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "flight_id": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "planned_path_points": [
      {
        "height": 123.234,
        "latitude": 13.23,
        "longitude": 123.234
      }
    ],
    "remaining_distance": 0,
    "remaining_time": 0,
    "result": 0,
    "status": "wayline_ok",
    "track_id": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "way_point_index": 1
  },
  "need_reply": 1,
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 16540709686556,
  "method": "takeoff_to_point_progress"
}
```

DRC 链路状态通知（已废弃）

注意：本协议不再维护且不准确，通过机场上报的设备属性“drc_state”或使用“DRC-心跳”可感知更准确的DRC链路状态。

Topic: thing/product/{gateway_sn}/events

Direction: up

Method: drc_status_notify

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误
drc_state	DRC 状态	enum_int	{"0":"未连接","1":"连接中","2":"已连接"}	

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "drc_state": 2,
    "result": 0
  },
  "need_reply": 1,
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "drc_status_notify"
}

```

DRC-飞行控制无效原因通知

DRC-飞行控制是一个无人机综合控制功能，若不可用，则无法使用飞行控制能力

Topic: thing/product/{gateway_sn}/events

Direction: up

Method: joystick_invalid_notify

Data:

Column	Name	Type	constraint	Description
reason	任务状态	int	{ "0": "遥控器失联", "1": "低电量返航", "2": "低电量降落", "3": "靠近限飞区", "4": "遥控器夺权（例如：触发了返航，B控夺权）" }	

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "reason": 0
  },
  "need_reply": 1,
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "joystick_invalid_notify"
}

```

上报拍照进度

当拍照行为需持续进行时会通过该事件上报进度。目前仅支持：全景拍照模式

Topic: thing/product/{gateway_sn}/events

Direction: up

Method: camera_photo_take_progress

Data:

Column	Name	Type	constraint	D
output	输出	struct		
»status	拍照状态	enum_string	{"fail":"失败","in_progress":"执行中","ok":"完成"}	
»progress	进度	struct		
»»current_step	执行步骤	enum_int	{"3000":"全景图拍摄未开始或者已结束","3002":"全景图正在拍摄","3005":"全景图合成中"}	
»»percent	进度值	int	{"max":"100","min":"0","step":"1"}	
»ext	扩展内容	struct		
»»camera_mode	当前相机模式	enum_int	{"3":"全景拍照"}	
result	返回码	int		非误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "output": {
```

```
        "progress": {
            "current_step": 0,
            "percent": 100
        },
        "status": "ok"
    },
    "result": 0
},
"need_reply": 1,
"tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
"timestamp": 1654070968655,
"method": "camera_photo_take_progress"
}
```

Service

飞行控制权抢夺

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: flight_authority_grab

Data: null

Example:

```
{
    "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "data": {},
    "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "timestamp": 1654070968655,
    "method": "flight_authority_grab"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: flight_authority_grab

Data:

result	返回码	int		非 0 代表错误
--------	-----	-----	--	----------

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "flight_authority_grab"
}

```

负载控制权抢夺

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: payload_authority_grab

Data:

Column	Name	Type	constraint	Description
payload_index	负载枚举值	text		镜头负载与挂载位置枚举值。非标准的 device_mode_key，格式为 {type-subtype-gimbalindex}，可以参考产品支持

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "payload_index": "39-0-7"
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "payload_authority_grab"
}

```

Direction: up

Method: payload_authority_grab

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "payload_authority_grab"
}

```

进入指令飞行控制模式

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: drc_mode_enter

Data:

Column	Name	Type	constraint	Descri
mqtt_broker	Broker 连接 信息	struct		获取 MQ 服务的地址 证信息
»address	服务器连 接地 址	text		服务器连 址，例如 192.0.2.1: mqtt.dji.c

»client_id	客户端 ID	text		可自定义 MQTT 客户端 ID。建议准备的 SN 码以与具有前缀组合如，drc-4
»username	用户名	text		建立连接的用户名
»password	密码	text		建立连接所需要的
»expire_time	认证信息过期时间	int	{"unit_name": "秒 / s"}	在有效期信息可以用，另外息过期后会影响已接的设备
»enable_tls	是否启用 TLS	bool		启用 TLS MQTT 链加密
osd_frequency	OSD 频率	int	{"max":30,"min":1,"unit_name": "赫兹 / Hz"}	设置 OSD 率
hsi_frequency	HSI 频率	int	{"max":30,"min":1,"unit_name": "赫兹 / Hz"}	设置 HSI 率

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "hsi_frequency": 1,
    "mqtt_broker": {
      "address": "mqtt.dji.com:8883",
      "client_id": "sn_a",
      "enable_tls": true,
      "expire_time": 1672744922,
      "password": "jwt_token",
      "username": "sn_a_username"
    },
    "osd_frequency": 10
  }
}

```

```
}  
  "method": "drc_mode_enter"
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: drc_mode_enter

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{  
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
  "data": {  
    "result": 0  
  },  
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
  "timestamp": 1654070968655,  
  "method": "drc_mode_enter"  
}
```

退出指令飞行控制模式

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: drc_mode_exit

Data: null

Example:

```
{  
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
  "data": {},  
}
```

}

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: drc_mode_exit

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "drc_mode_exit"
}

```

一键起飞

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: takeoff_to_point

Data:

Column	Name	Type	constraint
target_latitude	目标点纬度	double	{"max":90,"min":-90}

target_longitude	目标点经度	double	{"max":180,"min":-180}
target_height	目标点高度	float	{"max":1500,"min":2,"step":0.1 / m"}
security_takeoff_height	安全起飞高度	float	{"max":1500,"min":20,"step":0.1 / m"}
rth_mode	【必填】返航模式设置值	enum_int	{"0":"智能高度","1":"设定高度"

rth_altitude	返航高度	int	{"max":1500,"min":2,"step":1,m"}
rc_lost_action	遥控器失控动作	enum_int	{"0":"悬停","1":"着陆(降落)","
commander_mode_lost_action	【必填】指点飞行失控动作	enum_int	{"0":"继续执行指点飞行任务'任务，执行普通失控行为"}
commander_flight_mode	【必填】指点飞行模式设置值	enum_int	{"0":"智能高度飞行","1":"设定
commander_flight_height	【必填】指点飞行高度	float	{"max":3000,"min":2,"step":0./ m"}
flight_id	一键起飞任务UUID	text	
max_speed	一键起飞的飞行过	int	{"max":15,"min":1,"unit_name

	能达到的最大速度		
simulate_mission	是否在模拟器中执行任务	struct	
»is_enable	是否开启模拟器任务	enum_int	{"0":"不开启","1":"开启"}
»latitude	纬度	double	{"max":"90.0","min":"-90.0"}
»longitude	经度	double	{"max":"180.0","min":"-180.0"}
flight_safety_advance_check	飞行安全预检查	bool	{"0":"关闭","1":"开启"}

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "flight_id": "ABDEAC21DCADDA",
    "max_speed": 12,
    "rc_lost_action": 0,
    "rth_altitude": 100,
    "security_takeoff_height": 100,
    "target_height": 100,
    "target_latitude": 12.23,
    "target_longitude": 12.32,
    "commander_mode_lost_action": 1,
    "commander_flight_height": 80,
    "flight_safety_advance_check": 1
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "takeoff_to_point"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: takeoff_to_point

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
```

flyto 飞向目标点

特别说明：飞行器有最低飞行高度(20m)安全保障机制，如果飞行器相对起飞点高度低于20m，会先上升到20m

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: fly_to_point

Data:

Column	Name	Type	constraint
fly_to_id	飞向目标点 ID	text	
max_speed	flyto 的飞行过程中能达到的最大速度	int	{"max":15,"min":0,"unit_name":"米每秒 / m/s"}
points	flyto 目标点列表	array	{"size": -, "item_type": struct}
»latitude	目标点纬度	double	{"max":90,"min":-90}

»longitude	目标点经度	double	{"max":180,"min":-180}
»height	目标点高度	float	{"max":10000,"min":2,"step":0.1,"unit_name":"米 / m"}

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "fly_to_id": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "max_speed": 12,
    "points": [
      {
        "height": 100,
        "latitude": 12.23,
        "longitude": 12.23
      }
    ]
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "fly_to_point"
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: fly_to_point

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "fly_to_point"
}
```

结束 flyto 飞向目标点任务

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: fly_to_point_stop

Data: null

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {},
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "fly_to_point_stop"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: fly_to_point_stop

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

    "data": {
      "result": 0
    },
    "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "timestamp": 1654070968655,
    "method": "fly_to_point_stop"
  }

```

更新 flyto 目标点

「一键起飞」或「flyto 飞向目标点」的过程中，可以通过该命令快速更新目标点

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: fly_to_point_update

Data:

Column	Name	Type	constraint
max_speed	指令飞行过程中能达到的最大速度	int	{"max":15,"min":1,"unit_name":"米每秒 / m/s"}
points	更新的目标点列表	array	{"size": -, "item_type": struct}
»latitude	目标点纬度	double	{"max":90,"min":-90}

»longitude	目标点经度	double	{"max":180,"min":-180}
»height	目标点高度	float	{"max":10000,"min":2,"step":0.1,"unit_name":"米 / m"}

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "max_speed": 12,
    "points": [
      {
        "height": 100,
        "latitude": 12.23,
        "longitude": 12.23
      }
    ]
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "fly_to_point_update"
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: fly_to_point_update



Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "fly_to_point_update"
}

```

负载控制-框选变焦

框选变焦功能为在相机镜头的视野范围内，框选镜头中的目标区域，将会自动把画面切过去自动变焦+转云台

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_frame_zoom

Data:

Column	Name	Type	constraint
payload_index	相机枚举	text	
camera_type	相机类型	enum_string	{"ir":"红外","wide":"广角","zoom":"变焦"}
locked	机头和云台的相对关系是否锁定	bool	{"0":"仅云台转，机身不转","1":"锁定机头，身一起转"}
x	目标框左上角点坐标x	float	{"max":"1","min":"0","step":"0.000001","unit":"/"}

y	目标框左上角点坐标y	float	{"max":"1","min":"0","step":"0.000001","unit":"mm"}
width	目标框宽度	float	{"max":"1","min":"0","step":"0.000001","unit":"mm"}
height	目标框高度	float	{"max":"1","min":"0","step":"0.000001","unit":"mm"}

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "camera_type": "zoom",
    "height": 0.2,
    "locked": true,
    "payload_index": "39-0-7",
    "width": 0.2,
    "x": 0.5,
    "y": 0.5
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_frame_zoom"
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_frame_zoom

Data:

Column	Name	Type	constraint	Description
result	返回码	int		


```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_frame_zoom"
}

```

负载控制—切换相机模式

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_mode_switch

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的device_mode_key，格式为 {type-subtype-gimbalindex}，可以参考产品支持
camera_mode	相机模式	enum_int	{"0":"拍照","1":"录像","2":"智能低光","3":"全景拍照"}	

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "camera_mode": 0,
    "payload_index": "39-0-7"
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "timestamp": 1654070968655,

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_mode_switch

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_mode_switch"
}
```

负载控制—开始拍照

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_photo_take

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key，格式为 {type-subtype-gimbalindex}，可以参考产品支持 🔗

Example:

```

    "data": {
      "payload_index": "39-0-7"
    },
    "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "timestamp": 1654070968655,
    "method": "camera_photo_take"
  }

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_photo_take

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误
output	输出	struct		
»status	任务状态	enum_string	{"in_progress":"执行中"}	当全景拍照或其他持续性照行为时会上报状态信息表达后续会有持续的进度件上报，详细内容请查看 camera_photo_take_prog 事件

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_photo_take"
}

```

负载控制—停止拍照

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_photo_stop

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key, 格式为 {type-subtype-gimbalindex}, 可以参考产品支持 🔗

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "payload_index": "39-0-7"
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_photo_stop"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_photo_stop

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
```

```
    "timestamp": 1654070968655,  
    "method": "camera_photo_stop"  
}
```

负载控制—开始录像

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_recording_start

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的device_mode_key，格式为{type-subtype-gimbalindex}，可以参考产品支持 🔗

Example:

```
{  
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
  "data": {  
    "payload_index": "39-0-7"  
  },  
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
  "timestamp": 1654070968655,  
  "method": "camera_recording_start"  
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_recording_start

Data:

result	返回码	int		非 0 代表错误
--------	-----	-----	--	----------

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_recording_start"
}

```

负载控制—停止录像

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_recording_stop

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key, 格式为 {type-subtype-gimbalindex}, 可以参考产品支持

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "payload_index": "39-0-7"
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_recording_stop"
}

```

Direction: up

Method: camera_recording_stop

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_recording_stop"
}

```

负载控制—画面拖动控制

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_screen_drag

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key, 格式为 {type-subtype-gimbalindex}, 可以参考产品支持
locked	机头和云台的	bool	{"false": "仅云台转, 机身不	

	云台是否锁定		机头，云台和机身一起转"	
pitch_speed	云台 pitch 速度	double	{"unit_name": "度每秒 / degree/s"}	云台 pitch 速度
yaw_speed	云台 yaw 速度	double	{"unit_name": "度每秒 / degree/s"}	云台 yaw 速度，仅不锁机头时才生效

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "locked": true,
    "payload_index": "39-0-7",
    "pitch_speed": 0.1,
    "yaw_speed": 0.1
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_screen_drag"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_screen_drag

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  }
}
```



```
"method": "camera_screen_drag"
}
```

负载控制—双击成为 AIM

双击 aim 功能为在相机镜头的视野范围内，双击镜头中的目标点，该目标点将成为镜头视野的中心

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_aim

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key, 格式为 {type-subtype-gimbalindex}, 可以参考产品支持 🔗
camera_type	相机类型	enum_string	{"ir":"红外","wide":"广角","zoom":"变焦"}	
locked	机头和云台的相对关系是否锁定	bool	{"false":"仅云台转, 机身不转","true":"锁定机头, 云台和机身一起转"}	
x	目标坐标 x	double	{"max":1,"min":0}	目标坐标 x, 以镜头的左上角为坐标中心点, 水平方向为 x

y	目标坐标 y	double	{"max":1,"min":0}	目标坐标 y，以镜头的左上角为坐标中心点，竖直方向为 y
---	--------	--------	-------------------	------------------------------

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "camera_type": "zoom",
    "locked": true,
    "payload_index": "39-0-7",
    "x": 0.5,
    "y": 0.5
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_aim"
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_aim

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
}

```

负载控制—变焦

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_focal_length_set

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key 格式为 {type-subtype-gimbalindex}, 可参考产品支持
camera_type	相机类型	enum_string	{"ir":"红外","wide":"广角","zoom":"变焦"}	相机类型枚举
zoom_factor	变焦倍数	double	{"max":200,"min":2}	变焦倍数, 可见光是2-200, 红外是20

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "camera_type": "zoom",
    "payload_index": "39-0-7",
    "zoom_factor": 5
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_focal_length_set"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Method: camera_focal_length_set

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_focal_length_set"
}
```

负载控制—重置云台

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: gimbal_reset

Data:

Column	Name	Type	constraint	Description
payload_index	负载编号	text		负载编号，相机枚举值。非标准的 device_mode_key，格式为 {type-subtype-gimbalindex}，可以参考产品支持
reset_mode	重置模式类型	enum_int	{"0":"回中","1":"向下","2":"偏航回中","3":"俯仰向下"}	

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "payload_index": "39-0-7",
    "reset_mode": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "gimbal_reset"
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: gimbal_reset

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "gimbal_reset"
}

```

负载控制—Look At

lookat 功能指飞行器将从当前朝向转向实际经纬高度指定的点，在 M30/M30T 机型上建议使用锁定机头的方式，仅云台转动场景下在抵达云台限位角后 lookat 功能将出现异常

Topic: thing/product/{gateway_sn}/services

Direction: down

Data.

Column	Name	Type	constraint
payload_index	相机枚举	text	
locked	机头和云台的相对关系是否锁定	bool	{"false": "仅云台转，机身不转", "true": "锁定机头，云台和机身一起转"}
latitude	目标点纬度	double	{"max": 90, "min": -90}
longitude	目标点经度	double	{"max": 180, "min": -180}
height	目标点高度	float	{"max": 10000, "min": 2, "step": 0.1, "unit_name": "米 / m"}

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "height": 100,
    "latitude": 12.23,
    "locked": true,
    "longitude": 12.23,
    "payload_index": "39-0-7"
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_look_at

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_look_at"
}
```

负载控制—分屏

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_screen_split

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key，格式为 {type-subtype-gimbalindex}，可以参考产品支持 🔗

enable	是否使 能分屏	bool		开启还是关闭分屏
--------	------------	------	--	----------

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "enable": true,
    "payload_index": "39-0-7"
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_screen_split"
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_screen_split

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_screen_split"
}

```

负载控制—照片存储设置

Topic: thing/product/{gateway_sn}/services

Method: photo_storage_set

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key, 格式为 {type-subtype-gimbalindex}, 可以参考产品支持
photo_storage_settings	照片存储设置集合	array	{"size": -, "item_type": enum_string}	拍照存储类型 {current, vision, ir}, 可多选

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "payload_index": "39-0-7",
    "photo_storage_settings": [
      "current",
      "vision",
      "ir"
    ]
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "photo_storage_set"
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: photo_storage_set

Data:

result	返回码	int		非 0 代表错误
--------	-----	-----	--	----------

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "photo_storage_set"
}

```

负载控制—视频存储设置

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: video_storage_set

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key, 格式为 {type-subtype-gimbalindex}, 可以参考产品支持
video_storage_settings	视频存储设置集合	array	{ "size": -, "item_type": enum_string }	视频存储类型 {current, wide, zoom, ir}, 可多选

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",

```

```
        "current",
        "wide",
        "zoom",
        "ir"
    ]
},
"tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
"timestamp": 1654070968655,
"method": "video_storage_set"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: video_storage_set

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "video_storage_set"
}
```

负载控制—相机曝光模式设置

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_exposure_mode_set

Data:

payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key, 格式为 {type-subtype-gimbalindex}, 可以参考 产品支持
camera_type	相机类型	enum_string	{"wide": "广角", "zoom": "变焦"}	相机类型枚举
exposure_mode	曝光模式	enum_int	{"1": "自动", "2": "快门优先曝光", "3": "光圈优先曝光", "4": "手动曝光"}	

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "data": {
    "camera_type": "zoom",
    "exposure_mode": 1,
    "payload_index": "39-0-7"
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_exposure_mode_set"
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_exposure_mode_set

Data:

result	返回码	int		非 0 代表错误
--------	-----	-----	--	----------

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_exposure_mode_set"
}

```

负载控制—相机曝光值调节

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_exposure_set

Data:

Column	Name	Type	
payload_index	相机枚举	text	
camera_type	相机类型	enum_string	{"wide": "广角", "zoom": "变焦"}
exposure_value	曝光值	enum_string	{"1": "-5.0EV", "2": "-4.7EV", "3": "-4.3EV", "4": "-4.0EV", "5": "-3.7EV", "6": "-3.4EV", "7": "-3.1EV", "8": "-2.8EV", "9": "-2.5EV", "10": "-2.2EV", "11": "-1.9EV", "12": "-1.6EV", "13": "-1.3EV", "14": "-1.0EV", "15": "-0.7EV", "16": "-0.4EV", "17": "-0.1EV", "18": "0", "19": "0.3EV", "20": "0.6EV", "21": "0.9EV", "22": "1.2EV", "23": "1.5EV", "24": "1.8EV", "25": "2.1EV", "26": "2.4EV", "27": "2.7EV", "28": "3.0EV", "29": "3.3EV", "30": "3.6EV", "31": "3.9EV", "32": "4.2EV", "33": "4.5EV", "34": "4.8EV", "35": "5.1EV", "36": "5.4EV", "37": "5.7EV", "38": "6.0EV", "39": "6.3EV", "40": "6.6EV", "41": "6.9EV", "42": "7.2EV", "43": "7.5EV", "44": "7.8EV", "45": "8.1EV", "46": "8.4EV", "47": "8.7EV", "48": "9.0EV", "49": "9.3EV", "50": "9.6EV", "51": "9.9EV", "52": "10.2EV", "53": "10.5EV", "54": "10.8EV", "55": "11.1EV", "56": "11.4EV", "57": "11.7EV", "58": "12.0EV", "59": "12.3EV", "60": "12.6EV", "61": "12.9EV", "62": "13.2EV", "63": "13.5EV", "64": 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Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_exposure_set"
}
```

负载控制—相机对焦模式设置

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_focus_mode_set

Data:

payload_index	相机枚举	text		相机枚举的 device_ 格式为 subtype gimbal 参考产品
camera_type	相机类型	enum_string	{"wide": "广角", "zoom": "变焦"}	相机类型意: Mat 飞行器 zoom 该参数
focus_mode	对焦模式	enum_int	{"0": "MF", "1": "AFS", "2": "AFC"}	

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "camera_type": "zoom",
    "focus_mode": 1,
    "payload_index": "39-0-7"
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_focus_mode_set"
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_focus_mode_set

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

    "data": {
      "result": 0
    },
    "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "timestamp": 1654070968655,
    "method": "camera_focus_mode_set"
  }

```

负载控制—相机对焦值设置

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_focus_value_set

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key，格式为 {type-subtype-gimbalindex}，可以参考产品支持
camera_type	相机类型	enum_string	{"wide": "广角", "zoom": "变焦"}	相机类型枚举，注意：Matrice 30 系列飞行器只支持 zoom 镜头下配置该参数
focus_value	对焦值	int		对焦值，范围参见飞行器物模型属性 zoom_max_focus_value 和 zoom_min_focus_value

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "camera_type": "zoom",

```



```
    "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "timestamp": 1654070968655,
    "method": "camera_focus_value_set"
  }
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_focus_value_set

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_focus_value_set"
}
```

负载控制一点对焦

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: camera_point_focus_action

Data:

payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key, 格式为 {type-subtype-gimbalindex}, 可以参考 产品支持
camera_type	相机类型	enum_string	{"wide": "广角", "zoom": "变焦"}	相机类型枚举, 注意: Matrice 30 系列飞行器只支持 zoom 镜头下配置该参数
x	对焦点坐标 x	double	{"max": 1, "min": 0}	以镜头的左上角为坐标中心点, 水平方向为 x
y	对焦点坐标 y	double	{"max": 1, "min": 0}	以镜头的左上角为坐标中心点, 竖直方向为 y

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxx",
  "data": {
    "camera_type": "zoom",
    "payload_index": "39-0-7",
    "x": 0.5,
    "y": 0.5
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_point_focus_action"
}

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: camera_point_focus_action

Data:

result	返回码	int	非 0 代表错误
--------	-----	-----	----------

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "camera_point_focus_action"
}
```

负载控制—红外测温模式设置

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: ir_metering_mode_set

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key，格式为 {type-subtype-gimbalindex}，可以参考产品支持 🔗
mode	测温模式	enum_int	{"0": "关闭测温", "1": "点测温", "2": "区域测温"}	

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "mode": 1,
  }
}
```

```
"timestamp": 1654070968655,  
"method": "ir_metering_mode_set"  
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: ir_metering_mode_set

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{  
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
  "data": {  
    "result": 0  
  },  
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",  
  "timestamp": 1654070968655,  
  "method": "ir_metering_mode_set"  
}
```

负载控制—红外测温点设置

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: ir_metering_point_set

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key，格

				payload_index}, 可以参考产品支持
x	测温点坐标 x	double	{"max":1,"min":0}	以镜头的左上角为坐标中心点，水平方向为 x
y	测温点坐标 y	double	{"max":1,"min":0}	以镜头的左上角为坐标中心点，竖直方向为 y

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "payload_index": "39-0-7",
    "x": 0.5,
    "y": 0.5
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "ir_metering_point_set"
}
```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: ir_metering_point_set

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
}
```

```
}

```

负载控制—红外测温区域设置

Topic: thing/product/{gateway_sn}/services

Direction: down

Method: ir_metering_area_set

Data:

Column	Name	Type	constraint	Description
payload_index	相机枚举	text		相机枚举值。非标准的 device_mode_key，格式为 {type-subtype-gimbalindex}，可以参考产品支持
x	测温区域左上角点坐标 x	double	{"max":1,"min":0}	以镜头的左上角为坐标中心点，水平方向为 x
y	测温区域左上角点坐标 y	double	{"max":1,"min":0}	以镜头的左上角为坐标中心点，竖直方向为 y
width	测温区域宽度	double	{"max":1,"min":0}	
height	测温区域高度	double	{"max":1,"min":0}	

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "height": 0.5,
    "payload_index": "39-0-7",
    "width": 0.5,

```

```

    "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
    "timestamp": 1654070968655,
    "method": "ir_metering_area_set"
  }

```

Topic: thing/product/{gateway_sn}/services_reply

Direction: up

Method: ir_metering_area_set

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "result": 0
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1654070968655,
  "method": "ir_metering_area_set"
}

```

DRC

DRC-杆量控制

建立DRC链路之后，可通过“DRC-杆量控制”指令控制飞行器和云台姿态。发送频率需要保持5-10hz，才能比较精准地控制飞行器的运动。本协议无回包机制。

Topic: thing/product/{gateway_sn}/drc/down

Direction: down

Method: stick_control

Column	Name	Type	constraint	Description
roll	横滚通道	int	{"max": "1684", "min": "364", "unit_name": ""}	对应遥控器通道，控制器横滚（为左右平移）。杆围 [1024±6 1024为中（无动作数值增大向右倾斜小表示向斜。
pitch	俯仰通道	int	{"max": "1684", "min": "364", "unit_name": ""}	对应遥控器通道，控制器俯仰（为前后平移）。杆围 [1024±6 1024为中（无动作数值增大向前俯冲小表示向头。
throttle	升降通道	int	{"max": "1684", "min": "364", "unit_name": ""}	对应遥控器通道，控制器升降。 范围 [1024±6 1024为悬态，数值表示升高小表示降
yaw	偏航通道	int	{"max": "1684", "min": "364", "unit_name": ""}	对应遥控器通道，控制器偏航（为左右旋转）。杆围 [1024±6

				(无动作 数值增大 顺时针旋 减小表示 针旋转。
--	--	--	--	--------------------------------------

Example:

```

{
  "seq": 1,
  "method": "stick_control",
  "data": {
    "roll": 1024,
    "pitch": 1024,
    "throttle": 1024,
    "yaw": 1024,
    "gimbal_pitch": 1024
  }
}

```

DRC-飞行控制（已废弃）

注意：本协议已不再维护，使用“DRC-杆量控制”可获得更好的响应性能。

进入指令飞行模式后允许通过该指令控制飞行器航行方向与速度，发送的频率需要保持在5-10hz以内让设备能够比较精准的控制速度变化与方向。

Topic: thing/product/{gateway_sn}/drc/down

Direction: down

Method: drone_control

Data:

Column	Name	Type	constraint	Description
seq	命令序号	int		递增的序号，保证指令顺序执行。若 x、y、h、w 参数发生变

				要从 0 开始增长。
x	前进后退方向的速度	double	{"max":17,"min":-17,"unit_name":"米每秒 / m/s"}	前进后退的最大速度，负值表示向后移动
y	左右方向的速度	double	{"max":17,"min":-17,"unit_name":"米每秒 / m/s"}	左右移动的最大速度，负值表示向左移动
h	上下方向的速度	double	{"max":5,"min":-4,"unit_name":"米每秒 / m/s"}	向上向下移动的最大速度，负值表示向下移动
w	机身角速度	double	{"max":90,"min":-90,"unit_name":"度每秒 / degree/s"}	顺时针与逆时针的最大角速度，负值表示逆时针转动

Example:

```

{
  "data": {
    "h": 2.76,
    "seq": 1,
    "w": 2.86,
    "x": 2.34,
    "y": -2.45
  },
  "method": "drone_control"
}

```

Topic: thing/product/{gateway_sn}/drc/up

Direction: up

Method: drone_control

Column	Name	Type	constraint	Description
result	返回码	int		如果发送成功，则不会有返回码。 非 0 代表错误，异常case：没有飞行控制权，没有虚拟摇杆权限，数据包序号不对
output	输出	struct		
»seq	命令序号	int		递增的序号，保证指令顺序执行

Example:

```

{
  "data": {
    "output": {
      "seq": -1
    },
    "result": 319033
  },
  "method": "drone_control"
}
```

DRC-飞行器急停

Topic: thing/product/{gateway_sn}/drc/down

Direction: down

Method: drone_emergency_stop

Data: null

Example:

```

{
  "data": {},
  "method": "drone_emergency_stop"
}
```

Topic: thing/product/{gateway_sn}/drc/up

Method: drone_emergency_stop

Data:

Column	Name	Type	constraint	Description
result	返回码	int		非 0 代表错误

Example:

```

{
  "data": {
    "result": 0
  },
  "method": "drone_emergency_stop"
}

```

DRC-心跳

与 data 同级的 seq 是递增的序号，保证指令顺序执行

Topic: thing/product/{gateway_sn}/drc/down

Direction: down

Method: heart_beat

Data:

Column	Name	Type	constraint	Description
seq	【废弃】命令序号	int		递增的序号，保证指令顺序执行
timestamp	心跳发送时间戳	int	{"unit_name": "毫秒 / ms"}	业务方可以通过心跳协议判断 DRC 链路是否活跃，若一定期间未收到心跳回应可以判定 DRC 链路异常，对于 DRC 异常情况可以发起重试。若超过 1 分钟未收到业务方发送心跳协议，设备将会认为 DRC

链路。一旦链路异常，设备将退出 DRC 链路。

Example:

```
{
  "data": {
    "timestamp": 1670415891013
  },
  "method": "heart_beat",
  "seq": 1
}
```

Topic: thing/product/{gateway_sn}/drc/up

Direction: up

Method: heart_beat

Data:

Column	Name	Type	constraint	Description
seq	【废弃】命令序号	int		递增的序号，保证指令顺序执行
timestamp	心跳发送时间戳	int	{"unit_name": "毫秒 / ms"}	业务方可以通过心跳协议判断 DRC 链路是否活跃，若一定期间未收到心跳回应可以判定 DRC 链路异常，对于 DRC 异常情况可以发起重试。若超过 1 分钟未收到业务方发送心跳协议，设备将会认为 DRC 链路处于空闲退出 DRC 链路。

Example:

```
{
  "data": {
    "timestamp": 1670415891013
  },
  "method": "heart_beat",
}
```

DRC-避障信息上报

Topic: thing/product/{gateway_sn}/drc/up

Direction: up

Method: hsi_info_push

Data:

Column	Name	Type	constraint	Description
up_distance	上方的障碍物距离	int	{"unit_name": "毫米 / mm"}	
down_distance	下方的障碍物距离	int	{"unit_name": "毫米 / mm"}	
up_enable	上视避障开关状态	bool		
up_work	上视避障工作状态	bool		
down_enable	下视避障开关状态	bool		
down_work	下视避障工作状态	bool		
left_enable	左视避障	bool		

	左视避障开关状态			
left_work	左视避障工作状态	bool		
right_enable	右视避障开关状态	bool		
right_work	右视避障工作状态	bool		
front_enable	前视避障开关状态	bool		
front_work	前视避障工作状态	bool		
back_enable	后视避障开关状态	bool		
back_work	后视避障工作状态	bool		
vertical_enable	垂直避障开关状态	bool		
vertical_work	垂直避障工作状态	bool		

horizontal_enable	水平避障开关状态	bool		
horizontal_work	水平避障工作状态	bool		
around_distances	周边的障碍物距离	array	{ "size": 360, "item_type": int }	水平方向观察点，分布在[0,360)的角度区间，0对应机头方向正前方，顺时针分布，例如0度为机头正前方，90度为飞行器正右方。每个数值表示该角度上障碍物与飞行器距离，60000 表示该角度没有障碍物。若上报空数组，意味任意角度都无障碍物。若上报 4 个数据的数组，意味该数据为 TOF 避障数据，在红外避障失效时上报，譬如夜晚场景。

Example:

```

{
  "method": "hsi_info_push",
  "timestamp": 1670415891013,
  "data": {
    "up_distance": 10,
    "down_distance": 10,
    "around_distance": [
      10,
      8,
      9,
      16,
      2
    ],
    "up_enable": true,
    "up_work": true,
    "down_enable": true,
  }
}

```



```

        "right_enable": true,
        "right_work": true,
        "front_enable": true,
        "front_work": true,
        "back_enable": true,
        "back_work": true,
        "vertical_enable": true,
        "vertical_work": true,
        "horizontal_enable": true,
        "horizontal_work": true
    }
}

```

DRC-图传链路延时信息上报

Topic: thing/product/{gateway_sn}/drc/up

Direction: up

Method: delay_info_push

Data:

Column	Name	Type	constraint	Description
sdr_cmd_delay	图传协议命令链路延时	int	{"unit_name": "毫秒 / ms"}	图传协议命令链路延时
liveview_delay_list	图传视频码流延时	array	{"size": -, "item_type": struct}	图传视频码流延时，多路码流
»video_id	码流编号	text		码流编号
»liveview_delay_time	码流延时	int	{"unit_name": "毫秒 / ms"}	码流延时

Example:

```

{
    "data": {
        "liveview_delay_list": [

```

```

    },
    {
      "liveview_delay_time": 80,
      "video_id": "1581BN210004555439234/53-0-0/normal-0"
    }
  ],
  "sdr_cmd_delay": 10
},
"timestamp": 1670415891013
}

```

DRC-高频 osd 信息上报

Topic: thing/product/{gateway_sn}/drc/up

Direction: up

Method: osd_info_push

Data:

Column	Name	Type	constraint	Description
attitude_head	飞行器姿态 head 角	double	{"unit_name":"度 / °"}	飞行器姿态 head 角
latitude	飞行器纬度	double	{"unit_name":"度 / °"}	
longitude	飞行器经度	double	{"unit_name":"度 / °"}	
height	飞行器高度	double	{"unit_name":"度 / °"}	飞行器海拔高度
speed_x	当前飞行器 x 坐标方向的速度	double	{"unit_name":"米每秒 / m/s"}	当前飞行器 x 坐标方向的速度
speed_y	当前飞行器 y 坐标方向的速度	double	{"unit_name":"米每秒 / m/s"}	当前飞行器 y 坐标方向的速度

speed_z	当前飞行器 z 坐标方向的速度	double	{"unit_name":"米每秒 / m/s"}	
gimbal_pitch	云台 pitch 角	double	{"unit_name":"度 / °"}	
gimbal_roll	云台 roll 角	double	{"unit_name":"度 / °"}	
gimbal_yaw	云台 yaw 角	double	{"unit_name":"度 / °"}	

Example:

```
{
  "data": {
    "attitude_head": 60,
    "gimbal_pitch": 60,
    "gimbal_roll": 60,
    "gimbal_yaw": 60,
    "height": 10,
    "latitude": 10,
    "longitude": 10,
    "speed_x": 10,
    "speed_y": 10,
    "speed_z": 10
  },
  "timestamp": 1670415891013
}
```